

Effect of Bi₂O₃ on structural, optical and radiations shielding properties of glass system

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ABSTRACT

This research focuses on the synthesis and characterization of glass compositions containing Bi₂O₃ as a dopant for enhanced radiation shielding capabilities. The compositions, formulated as (80-x)B₂O₃ + 10Na₂O + 10BaCl₂ + xBi₂O₃ with x values of 0, 15, 30, and 45 mol%, were subjected to X-ray diffraction analysis to examine their structural properties. The X-ray diffraction analysis indicated that samples Bi0.0 and Bi0.15 exhibit an amorphous structure, while samples Bi0.30 and Bi0.45 display crystallization. Furthermore, the density of the samples was determined using the Archimedes method and the results revealed an increase from 2.5 to 5.4 g/cm³. The optical properties of the prepared glass samples were measured using UV–VIS spectrophotometer within the spectral range of 200–1200 nm. The efficiency of radiation shielding was evaluated through the Phy-x program software, providing insights into the attenuation of radiation across the different Bi₂O₃ doping levels. The findings illustrate a notable rise in the mass attenuation coefficient from 13.32 to 87.74 cm²/g correlating with the increased Bi₂O₃ content. Additionally, the effective atomic number exhibited an escalation from 30.75 to 76.96 cm²/g. The effective electron density demonstrated an increase in the glass samples, rising from 10.3 × 10²³ to 13.2 × 10²³ el/g for samples Bi0.0 and Bi0.15. In contrast, the effective electron density exhibited a decrease in the Bi0.30 and Bi0.45 samples, declining from 10.3 × 10²³ to 8.2 × 10²³ el/g. Simultaneously, the half-value layer witnessed a decrease from 0.021 to 0.001 cm, while the mean free path concurrently decreased from 0.030 to 0.002 cm. Moreover, the fast neutron removal cross section was examined, the results show the values fluctuating between 0.093 and 0.105 for glass samples.

1. Introduction

The utilization of glass and glass ceramic materials for radiation shielding applications has garnered significant attention in various fields, including nuclear technology, healthcare, and aerospace. The incorporation of dopants, such as Bi₂O₃, offers a promising avenue for enhancing the radiation attenuation properties of these materials. Bi₂O₃ dopants enhance radiation attenuation properties and improves absorption and scattering of ionizing radiation essential for worker safety in nuclear facilities and patient protection in healthcare settings. Bi₂O₃

incorporation leads to innovative radiation shielding technology advancements and as a promising dopant for improving radiation shielding effectiveness in diverse applications. Glass & glass ceramic materials doped with Bi₂O₃ offer potential for safe and effective radiation shielding solutions (Al-Fakeh et al., 2023; Algradee et al., 2022; Murshed et al., 2024; Saleh et al., 2022; Taya et al., 2024).

Boric oxide (B₂O₃) assumes a pivotal role as a compelling glass former and desirable material, distinguished for its distinctive characteristics. Borate glasses exhibit exceptional properties such as a low melting point, superior glass-forming capacity, thermal stability, and

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outstanding optical transparency. The versatile structures of borate glasses find wide-ranging applications across diverse sectors including optical glass manufacturing, opto-acoustical electronics, solid-state laser development, piezoelectric actuators, microelectronics, and luminescent material production. Furthermore, the introduction of various transition metals, alkaline earth elements, and alkali metals into borate-based glass compositions allows for the exploration of their linear and nonlinear optical properties, rendering them promising candidates for a spectrum of technological applications. Moreover, borate glasses exhibit heightened reactivity in aqueous environments compared to other glass types, showcasing rapid response characteristics that uniquely position them for a myriad of biomedical applications necessitating precise ion release control (Algrade et al., 2022; Saleh et al., 2021; Algrade et al., 2021; Saleh et al., 2021; Algrade et al., 2021; Abdelwhab et al., xxxx).

The addition of BaCl_2 to borate glass can lead to a range of modifications in the physical, chemical, and optical properties of the glass, offering opportunities to tailor its properties for specific applications. BaCl_2 addition can alter the density and refractive index of the borate glass, influencing optical properties such as transparency and light transmission. Additionally, the presence of BaCl_2 can induce structural modifications in the borate glass network, potentially affecting bonding configurations and overall stability. Furthermore, BaCl_2 may affect the thermal expansion coefficient and temperature stability of the glass, impacting its performance in high-temperature applications (Brekhovskikh et al., 2010). On the other hand, Na_2O acts as a network modifier in borate glass, disrupting the boron oxide network structure and altering the glass properties. The addition of Na_2O can affect the refractive index of the glass, influencing its optical properties and light transmission characteristics. Also, Na_2O can facilitate ion exchange processes in the glass, leading to changes in surface properties and potentially enhancing certain functionalities (Brekhovskikh et al., 2010; Ezhil PavaiP and Dharsini, 2017).

Various materials, including polymers, glass, and other diverse compositions, are extensively utilized as shields against gamma rays due to their unique properties and effectiveness in attenuating radiation. Polymers play a significant role in gamma ray shielding due to their versatile characteristics and customizable properties. Polymers, with their flexibility, lightweight nature, and ease of shaping, offer a cost-effective and practical solution for gamma ray attenuation in various applications. By incorporating specific additives or fillers, polymers can be tailored to enhance their shielding effectiveness against gamma rays. Lead-free polymer composites, utilizing materials such as boron, barium, or tungsten, are particularly popular for their ability to absorb and scatter gamma rays effectively. Additionally, polymers provide a convenient option for creating protective barriers in environments where flexibility, portability, and ease of installation are key considerations (Alharshan Gharam et al., 2020; Gouda et al., 2023; Obeid et al., 2022; Osman et al., 2023; Alsayed et al., 2021).

The study objective to synthesize and characterize compositions with varying Bi_2O_3 levels. The determination of the structure and optical properties of the glass system was conducted. Also, the research intends to evaluate the effectiveness of radiation shielding across different Bi_2O_3 doping levels using Phy/x program software, with the ultimate goal of optimizing the radiation shielding performance of the glass and glass ceramic samples.

2. Methodology and materials

2.1. Preparation and analysis methods

A synthetic borate glass was synthesized by introducing specific elemental oxides and produced through the melt-quenching process. The glass composition followed the equation $(80-x)\text{B}_2\text{O}_3 + 10\text{Na}_2\text{O} + 10\text{BaCl}_2 + x\text{Bi}_2\text{O}_3$, where x values are 0, 15, 30, and 45 mol%, were employed to formulate a series of glass samples with varying Bi_2O_3 concentrations. Raw materials were blended, pulverized, and

subsequently placed in porcelain crucibles before being subjected to heating in an electric furnace at 1000°C for 0.5 h. The samples were stirred every ten minutes in order to prevent the formation of bubbles within the samples. The molten samples were then cast into molds and promptly transferred to an annealing furnace set at 250°C for 1.5 h, followed by a gradual cooling process to room temperature upon furnace shutdown. Subsequently, the glass specimens were stored in the dried furnace for a day to prevent moisture absorption. The samples are labeled as Bi0.0 , Bi0.15 , Bi0.30 , and Bi0.45 based on the Bi_2O_3 content. The photograph illustrating the prepared samples is presented in Fig. 1.

2.2. Structural phase investigation

To investigate the structural variations in the samples with varying concentrations of Bi_2O_3 in samples, the X-ray diffraction (XRD) patterns of the samples were meticulously examined. The XRD data was acquired utilizing a Bruker D8 Advance XRD Pan Analytic Model X-ray diffractometer, encompassing diffraction angles ranging from 10 to 90 degrees.

2.3. Density and molar volume analysis

The density of the samples was determined using Archimedes' principle with toluene as the immersion liquid. The relationship used for density calculation is (Saleh et al., 2021):

$$\rho = \frac{w_a}{w_a - w_t} \times \rho_t \quad (1)$$

where w_a is the weight of the sample in air, w_t is the weight of the sample in toluene, and ρ_t is the density of toluene. The densities of the samples are presented in Table 1. The molar volume of the samples was calculated using the formula (Algrade et al., 2021):

$$V_m = \frac{\text{Molecularweightoftheglasssample}}{\rho_g} \quad (2)$$

where V_m represents the molar volume, and ρ_g denotes the density of the glass sample. The density increased with the rise in Bi_2O_3 content in the glass systems, attributed to the replacement of lighter oxides (B_2O_3) by the heavier metal oxide (Bi_2O_3), as shown in Fig. 2. Additionally, Fig. 2 illustrates that the molar volume of the glass samples also increases with the augmentation of Bi_2O_3 content in the glass systems.

2.4. Optical properties

The absorption (A) spectra were measured using a spectrophotometer Jasco V-570 Corp., in the range of wavelengths, 200 – 2000 nm with a spectral resolution of ± 2 nm.

3. Gamma-Ray attenuation properties

The calculation of nuclear radiation shielding parameters such as mass attenuation coefficients (MAC), half-value layer (HVL), effective atomic number (N_{eff}) and others were performed using the user-friendly online Photon Shielding and Dosimetry (PSD) software, which offers advanced tools for precise dosimetry calculations and photon shielding analysis. This specialized software, accessible at <https://phy-x.net/PSD> (Sakar et al., 2020), provides an intuitive interface for researchers to efficiently evaluate and optimize radiation shielding strategies in various applications. In addition to the theoretical calculations, the MAC values for the prepared glass samples were determined experimentally using Cs-137 and Co-60 point sources at photon energies of 0.661, 1.171, and 1.333 MeV. The experimental setup employed 2×2 in. thallium-doped sodium iodide $[\text{NaI}(\text{Tl})]$ scintillation detectors, which are widely used for gamma spectroscopy due to their high detection efficiency and good energy resolution. The typical energy resolution of the $\text{NaI}(\text{Tl})$ detector is approximately 7 % at 0.662 MeV (Cs-137) and

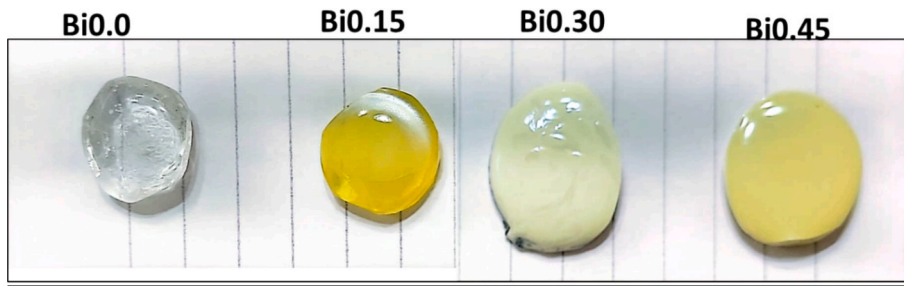


Fig. 1. Photographic image of the prepared glass samples.

Table 1

The densities and average molecular weight of the prepared samples.

Sample	Density (g/cm ³)	AMW (Average molecular weight) g/mol
Bi 0.0	2.47 ± 0.17	82.72 ± 4.14
Bi 0.15	4.16 ± 0.29	142.17 ± 7.10
Bi 0.30	5.17 ± 0.36	201.62 ± 10.10
Bi 0.45	5.90 ± 0.41	261.07 ± 13.05

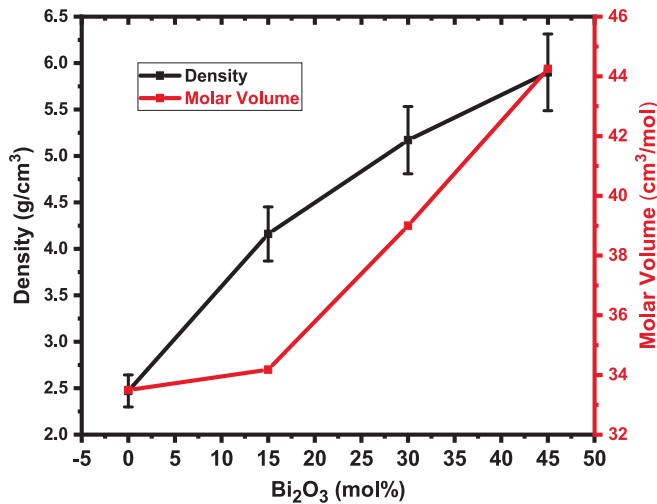


Fig. 2. Density and molar volume of all samples.

slightly varies with higher energies, providing sufficient resolution to distinguish the gamma photon energies of interest. This detector configuration ensured accurate and reliable measurement of gamma photon attenuation through the samples, complementing the theoretical predictions for the MAC evaluation. All experiments were conducted in triplicate to ensure reproducibility. The reported data represent the mean values. Confidence intervals were calculated using data from three independent experimental replicates per measurement. A confidence level of 95 % was applied in all cases.

3.1. Mass attenuation coefficient (MAC)

The mass attenuation coefficient (MAC) is a parameter that known as the linear attenuation coefficient of materials per unit mass, reflecting their ability to interact with incident radiation. It used as a fundamental property influencing the material's capacity to absorb, scatter, and attenuate radiation. The MAC is determined by the ratio of the linear attenuation coefficient (μ) to the material density (ρ), and it plays a role in assessing the material's effectiveness in attenuating radiation across various energy levels. The following formula (Taya et al., xxxx; Algradee

et al., 2021) yields the calculation of the total mass attenuation coefficients, with the material's thickness (x) and density (ρ).

$$MAC = \frac{\mu}{\rho} = \frac{1}{\rho x} \ln \frac{I_0}{I} \quad (3)$$

4. Role of Z_{eff} and N_{el} in photon interaction

The effective atomic number (Z_{eff}) and effective electron density (N_{el}) play crucial roles in understanding and characterizing the interaction of photons with matter, particularly in the context of radiation shielding and attenuation. The calculation methods for Z_{eff} and N_{el} are obtained through Equations (4) and (5) correspondingly. Z_{eff} denotes the effective atomic number, representing the quantity of electrons per atom that engage with photons. On the other hand, N_{el} signifies electron density, indicating the electron count per unit mass that interacts with photons and diminishes intensity as it traverses the material (Gunha and Al Huwayz, in press).

$$Z_{eff} = \sigma_{t,a} / \sigma_{t,el} \quad (4)$$

$$N_{el} = \mu_m / \sigma_{t,el} \quad (5)$$

where σ_{ta} is the total atomic cross-section and σ_{tel} electronic cross-section.

4.1. Half-Value layer (HVL) and mean free path (MFP)

The half-value layer (HVL) is determined using equation (6) as follows (Algradee et al., 2021):

$$HVL = 0.693 / \mu \quad (6)$$

where the HVL signifies the thickness of the material that reduces the radiation intensity by half. This characteristic parameter offers insights into the material's ability to attenuate ionizing radiation effectively.

The mean free path is the average distance a particle travels before undergoing a collision or interaction within a material. The formula for calculating the mean free path (MFP) is given by (Saleh et al., 2021):

$$MFP = 1 / \mu \quad (7)$$

where μ represents the linear attenuation coefficient.

4.2. Fast neutron absorption characteristics (FNAC)

The fast neutron absorption characteristics (FNAC) indicates a material's capacity to interact with and decelerate fast neutrons. It can be determined by computing the mass removal cross-section ($\Sigma R / \rho$) of individual elements in the material, as denoted in the following equation (Algradee et al., 2021):

$$FNAC = \Sigma R / \rho \Sigma \rho_i \quad (8)$$

where ρ_i denotes the partial density of the respective oxides within the material composition.

5. Results and discussion

5.1. Density and molar volume

The sample compositions along with their corresponding density and molar volume are detailed in Table 1. Observations indicated a direct correlation between the density of the samples and the Bi_2O_3 concentration. This observation attributed to the substitution of Bi_2O_3 atoms with higher molecular weights by B_2O_3 . Also, the molar volume exhibits an increase in the presence of higher Bi_2O_3 contents, as noted in the study by Saleh et al., (Saleh et al., 2021), as shown in Fig. 2. The experimental error in the determination of the density of the prepared glass samples reported in Table 1, is primarily due to several contributing factors. These include uncertainties in the weight measurement caused by the precision and calibration limits of the analytical balance used, as well as inaccuracies in volume determination arising from the measurement techniques employed. Additionally, handling and environmental conditions, such as fluctuations in temperature and humidity during measurement, may affect the accuracy. Collectively, these factors contribute to the overall measurement uncertainty, which is reflected in the reported error rate. Recognizing these sources of error is essential for a proper interpretation of the density results and further improvements in measurement protocols.

5.2. Structural characterization through XRD analysis

X-ray diffraction (XRD) analysis of the prepared glass samples revealed a random glass structure at low Bi_2O_3 concentrations. However, as the Bismuth oxide content increased to 30 and 45 mol%, three distinct crystalline phases emerged within the glass matrix: BaBiO_2Cl , Ba_2Bi , and BiCl_3 (Fig. 3). These phases were identified by comparing the XRD patterns with the JCPDS reference cards (75-0953), (71-0403), and (71-0666), respectively. The presence of these crystalline phases suggests the occurrence of partial crystallization or devitrification during the cooling process of the glass melt, leading to the formation and growth of these specific compounds as a result of potential phase separation in the glass composition.

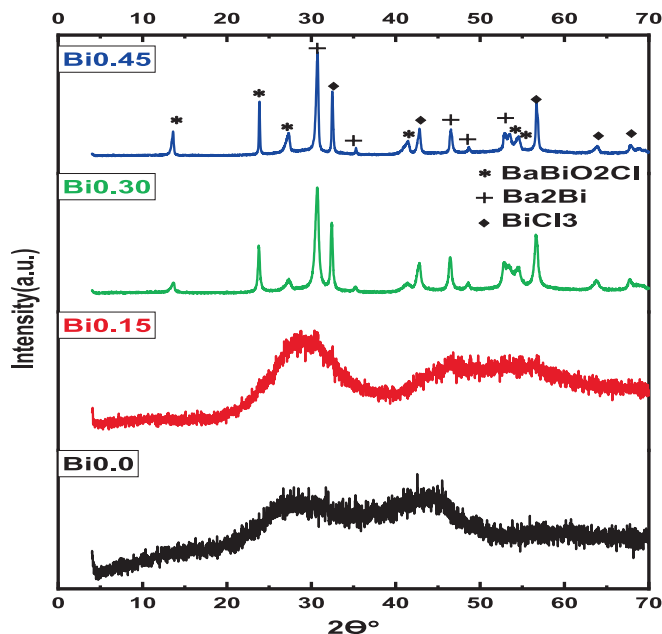


Fig. 3. The Structural Properties of the Prepared Samples (XRD Pattern).

5.3. Optical band gap and refractive index

As seen in Fig. 4, absorption spectra were carried out on all glass samples in the wavelength range of 200 to 2000 nm. Using the Davis and Mott relation (Mott and Davis, 1979); $\alpha(\nu)h\nu = [B(h\nu - E_{\text{opt}})]^s$, where the exponents may be 2 or 1/2 for an allowed indirect transition and an allowed direct transition, respectively, the optical band gap, E_{opt} , for all glass samples was calculated. where $\alpha(\nu)$ is the optical absorption coefficient and B is constant. The intersection of the y-axis's straight line, which represents $(\alpha h\nu)^{0.5}$, with the x-axis $h\nu$ when $(\alpha h\nu)^{0.5}$ equals zero (Tauc's method), as seen in Fig. 5, was used to determine the E_{opt} for an allowed indirect transition. The values were found to be 2.40, 2.20, 2.14 and 2.08 eV for the Bi0.00, Bi0.15, Bi0.30 and Bi0.45 samples, respectively as shown in Fig. 6 and presented in Table 2. According to the results, the E_{opt} decreases with increasing the Bi_2O_3 content due to the rise in the non-bridging oxygen (NBO) ions (D'Souza et al., 2020). The NBO ions contribute to the valence band maximum (VBM) and the non-bonding orbitals have larger energies than the bonding orbitals when a metal-oxygen link is broken, releasing the bond energy. This implies that the E_{opt} is reduced as a result of the shifting of VBM to higher energies caused by an increase in NBO ion concentration (Rani et al., 2014). Also, the observed decrease in the E_{opt} is related to the weaker bond strength of the Bi-O (80.3 kcal/mol) compared to that of B-O (192.7 kcal/mol) (Yasser et al., 2008).

The refractive index (n) for all samples was calculated based on E_{opt} value by the following eq. (9) (Algrade et al., 2017):

$$n = \left[\frac{3 - 2 \left(\frac{E_{\text{opt}}}{20} \right)^{0.5}}{\left(\frac{E_{\text{opt}}}{20} \right)^{0.5}} \right]^{0.5} \quad (9)$$

The n values for all samples were collected in Table 2 and plotted as a function of Bi_2O_3 in Fig. 6. The increasing value of n with an increase of Bi_2O_3 content may be due the higher polarizability of Bi^{3+} ions ($1.508 \text{ \AA}^3$) as compared to B^{3+} ions ($0.002 \text{ \AA}^3$) (Dimitrov and Komatsu, 2010).

6. Characterization of radiation shielding capabilities

6.1. Mass attenuation coefficient analysis

The mass attenuation coefficients (MAC) of the prepared glass samples Bi0.0, Bi0.15, Bi0.30, and Bi0.45 were determined experimentally

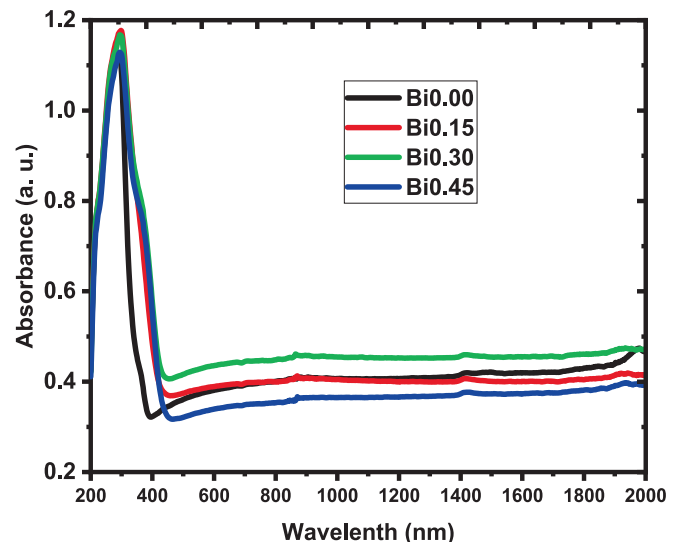


Fig. 4. UV-Vis absorption spectra of the glass samples.

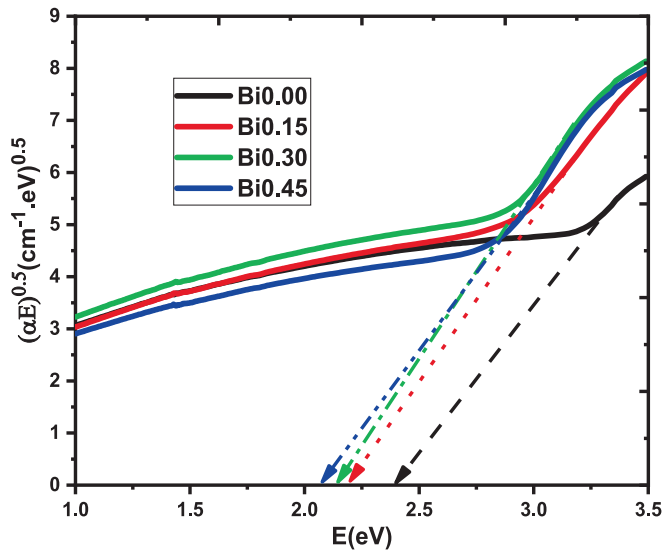


Fig. 5. The variation of $(\alpha B)^{0.5}$ versus energy for the samples.

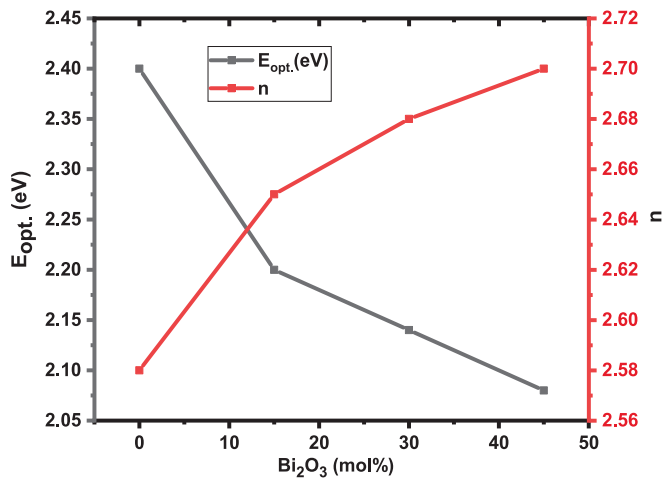


Fig. 6. The variation of $E_{opt.}$ and n versus Bi_2O_3 for the samples.

Table 2

Indirect optical band gap by Tauc's method $E_{opt.}$ and refractive index n , for glass samples.

Sample code	$E_{opt.}$ (eV)	n
Bi0.00	2.40	2.58
Bi0.15	2.20	2.65
Bi0.30	2.14	2.68
Bi0.45	2.08	2.70

and compared with theoretical values calculated using Phy-x software at photon energies of 0.661, 1.173, and 1.333 MeV. The results are presented in Table 3. The results show good agreement between theoretical

Table 3

Experimental determination of MAC (cm^2/g) of prepared samples compared with theoretical calculation by Phy-X software.

Energy (MeV)	Bi0.0			Bi0.15			Bi0.30			Bi0.45		
	Exp.	Phy-x	Div.	Exp.	Phy-x	Div.	Exp.	Phy-x	Div.	Exp.	Phy-x	Div.
0.661	0.071 ± 0.005	0.076	6.58%	0.087 ± 0.006	0.092	5.43%	0.095 ± 0.006	0.099	4.04%	0.106 ± 0.007	0.102	3.92%
1.173	0.065 ± 0.004	0.061	6.56%	0.062 ± 0.003	0.059	5.08%	0.064 ± 0.003	0.060	6.67%	0.065 ± 0.003	0.061	6.56%
1.333	0.051 ± 0.002	0.053	3.77%	0.059 ± 0.002	0.055	7.27%	0.057 ± 0.003	0.055	3.64%	0.059 ± 0.002	0.056	5.36%

Note: Each MAC value and confidence interval was derived from three independent measurements.

and experimental values for all samples across the measured energy range. For the Bi0.45 sample, the theoretical MAC values were 0.102, 0.061, and 0.056 cm^2/g at 0.661, 1.173, and 1.333 MeV respectively, while the corresponding experimental values were slightly higher at 0.106, 0.065, and 0.059 cm^2/g , with deviations ranging between 0.004 to 0.005 cm^2/g . Similarly, Bi0.30 exhibited theoretical and experimental MAC values of 0.099 vs. 0.095, 0.060 vs. 0.064, and 0.055 vs. 0.057 cm^2/g , showing small deviations within 0.004 cm^2/g . The Bi0.15 and Bi0.0 samples followed the same trend with marginal differences between theoretical and experimental values, indicating a deviation generally less than 0.006 cm^2/g . The minor discrepancies observed may be attributed to a combination of random experimental uncertainties, such as measurement noise and instrumental factors, as well as possible systematic errors arising from slight compositional variations in the glass samples. To quantitatively assess the variability and reliability of the experimental data, confidence intervals were calculated and are presented in Table 3, providing a statistical measure of uncertainty around the reported values. Overall, the close match between Phy-x theoretical predictions and experimental measurements, within the calculated confidence intervals, confirms the reliability of the theoretical model for estimating mass attenuation coefficients in these bismuth-doped glass systems.

Fig. 7 illustrates the mass attenuation coefficient (MAC) values for the synthesis glass samples at an energy gamma ray ranging from 0.015 to 15 MeV. At the lowest energy of 0.015 MeV, the maximum MAC values for the Bi0.0, Bi0.15, Bi0.30, and Bi0.45 samples were 13.32, 58.87, 77.56, and 87.74 cm^2/g , respectively. The samples Bi0.45 exhibited the highest MAC value due to its excellent density, while Bi0.0 displayed the lowest MAC value.

From the Fig. 7, it is evident that the MAC values exhibited a sharp decreasing at lower energy levels for all samples. it can be noticed that there is a significant peak in MAC values at 90.5 keV, corresponding to the absorption k-edge of bismuth. Following, the MAC values gradually decrease until they reach their minimal points, indicating the prevalence

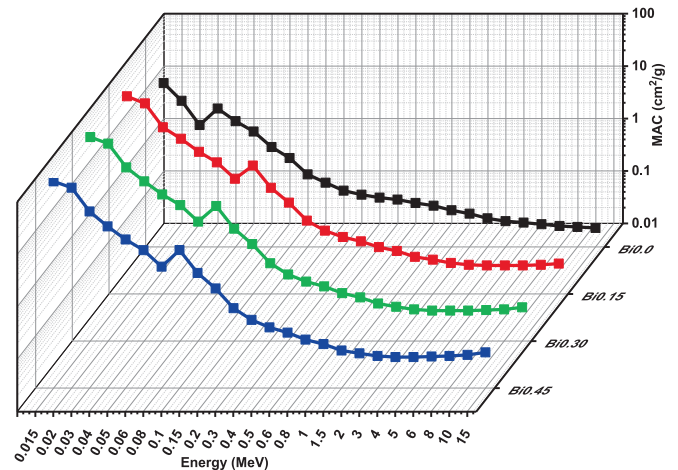


Fig. 7. Mass attenuation coefficient (MAC) values versus photon energy for the samples.

of the Compton effect wherein the cross-section is proportionate to the atomic number (Z) over the energy (E).

The dominance of the photoelectric effect is reflected in the low energy zone where $Z^5/E^{3.5}$, contributing to the pronounced decrease in MAC values. Following this, a decrease in MAC values is observed as the Compton effect interaction gains prominence. In the higher energy zone with the cross-section being proportionate to Z^2 and the logarithm of the photon energy (E), the values of MAC slightly increase.

Analyzing the data reveals that as the energy level increases, the mass attenuation coefficient generally decreases for all samples. This trend signifies the materials' ability to effectively reduce the intensity of gamma radiation as the energy levels rise, showcasing their inherent shielding capabilities.

Among the samples, Bi0.45 consistently demonstrates the highest mass attenuation coefficients across various energy levels, indicating its superior performance in attenuating gamma radiation compared to the other compositions. On the other hand, Bi0.0 consistently exhibits the lowest values among the samples, suggesting a lower efficiency in radiation shielding.

The influence of the bismuth oxide content on the mass attenuation properties of the prepared samples, with higher concentrations leading to enhanced radiation attenuation capacities. The findings highlight the importance of material composition in determining the effectiveness of radiation shielding, providing essential information for applications requiring radiation protection and safety measures in various fields. Furthermore, the mass attenuation coefficients determined in this study align well with the findings of three previous studies (Al-Fakeh et al., (Al-Fakeh et al., 2023); Saleh et al., (Saleh et al., 2021); Saleh et al., (Saleh et al., 2022)).

6.2. Effective atomic number (Z_{eff}) and electron density (N_{el})

The effective atomic numbers (Z_{eff}) derived using Phy-x software for the samples are detailed in Fig. 8. The findings reveal a pronounced increase in Z_{eff} with the increase of Bi_2O_3 content. Specifically, at an energy level of 0.015 MeV, the Z_{eff} values were approximately 30.75, 68.00, 74.34, and 76.96 for samples Bi0.0, Bi0.15, Bi0.30, and Bi0.45 respectively. High-energy photon interactions within materials are typically associated with Z_{eff} values and energy band. The energy interacts with the electron cloud surrounding the nucleus of the material as it traverses through it. Consequently, a higher Z_{eff} value can effectively attenuate the radiation intensity as it traverses the material. Sample Bi0.45 exhibit the top performer in radiation shielding effectiveness due to its superior Z_{eff} value.

Generally, higher Z_{eff} values indicate increased electron density in the material, which is essential for effective photon interaction and radiation attenuation. Observing the trends for each sample. Bi0.45

consistently shows the highest Z_{eff} values among the samples at various energy levels, indicating a higher electron density and effective photon interaction. This suggests that Bi0.45 offers superior radiation attenuation properties compared to the other compositions. Higher bismuth oxide content leads to enhanced Z_{eff} and photon interaction, resulting in improved radiation shielding properties.

The comparison of the effective atomic number (Z_{eff}) for the prepared glass sample Bi0.45 and commercial glasses (Rs-360, Rs 253 G18), ordinary concrete (OC), barium concrete (BaC), and other metals such as Cr reveals interesting insights and plotted in Fig. 9.

Initially, at lower energy levels, Bi0.45 exhibits relatively higher Z_{eff} values compared to the other materials, indicating its effectiveness in attenuating radiation at these energy levels. However, as the energy levels increase, the Z_{eff} values of the other materials also show notable performance, with some materials approaching the Z_{eff} values of Bi0.45 at higher energies.

Among the commercial glasses and concrete samples, variations in Z_{eff} values are observed, with certain materials showing competitive Z_{eff} values comparable to Bi0.45 at specific energy levels.

In comparison to the other materials like Cr and commercial glasses (Rs-360, Rs 253 G18), Bi0.45 maintains relatively higher Z_{eff} values, emphasizing its effectiveness in attenuating radiation across a range of energy levels. The consistent performance of Bi0.45 suggests its suitability as a shielding material for gamma radiation protection in various applications.

The electron density (N_{el}) results, are displayed in Fig. 10. The N_{el} values for the glass samples Bi0.0, Bi0.15, Bi0.30 and Bi0.45 samples show a clear trend. Initially, at lower energy levels, Bi0.0 exhibits higher electron density values compared to the other samples, gradually decreasing as the energy levels increase. In contrast, Bi0.45 demonstrates a lower electron density at lower energy levels but maintains relatively higher values as the energy levels rise. This trend indicates a variation in electron density behavior across different samples and energy levels, reflecting the materials' distinct characteristics in response to gamma radiation.

At energy 0.015 MeV, glass samples Bi0.0 and Bi0.15 demonstrate an increase in N_{el} $(10.2\text{--}13.2) \times 10^{23}$ electron/g, whereas a distinct decrease is observed in samples Bi0.30 and Bi0.45 $(10.2\text{--}8.17) \times 10^{23}$ electron/g. Interestingly, N_{el} exhibits an inverse relationship with Z_{eff} , showcasing a contrasting pattern. N_{el} signifies the density of electrons that engage in interactions per unit mass. Higher N_{el} values play a vital role in photon interactions with the samples, influencing the material's overall photon interaction behavior. The N_{el} results are influenced by both the energy levels and the concentrations of Bi_2O_3 . Specifically, at lower energy levels, the glass samples showcased the highest N_{el} values, which gradually decreased as the energy levels increased. This

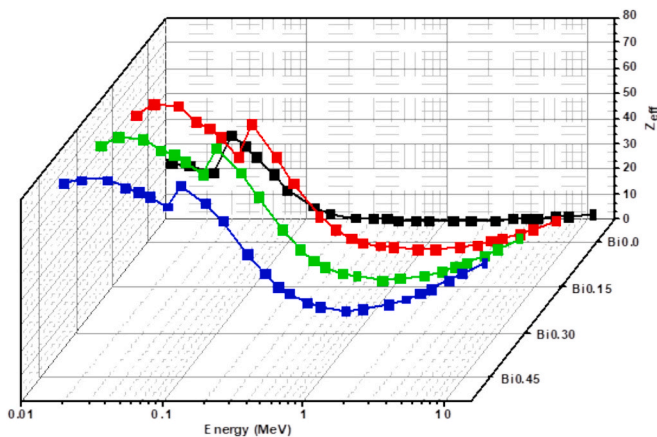


Fig. 8. Effective atomic numbers (Z_{eff}) versus photon energy of the samples.

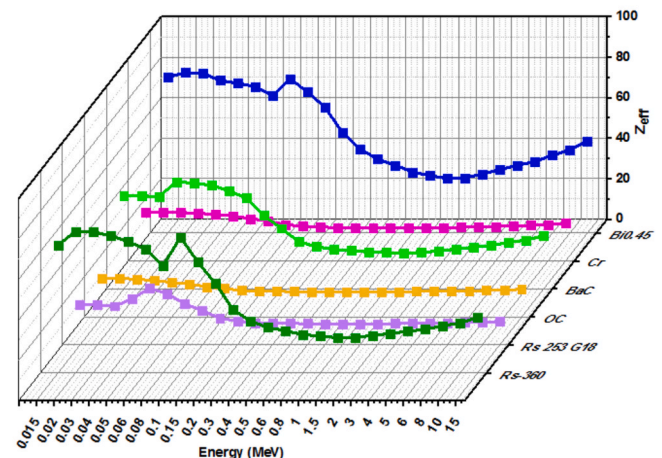


Fig. 9. A Comparison of the Z_{eff} for Bi0.45 sample with commercial materials.

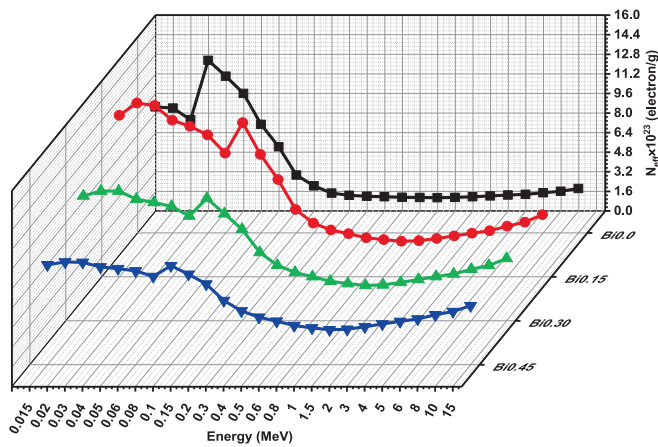


Fig. 10. Effective electron density (N_e) versus photon energy of the samples.

observation indicates that lower energy levels result in a greater likelihood of interaction.

Factors such as the distribution of electrons, the density and arrangement of atoms, can affect to relation between electron density and the atomic number. Therefore, while the effective atomic number and effective electron density are linked variables that contribute to radiation attenuation, the elaborate structure and structural characteristics of the glass material may lead to variations in how these properties evolve in response to changes in the elemental composition.

Additionally, As the Z_{eff} of the composite material increases due to the addition of heavy metal elements, the electron density per unit mass may decrease. This decrease in N_e can be attributed to the relative abundance of heavy atoms compared to lighter atoms with fewer electrons per atom.

In this case, the increased Z_{eff} lead to enhances the material's radiation attenuation capabilities by increasing photon interactions with the heavier atoms. At the same time, the decrease in N_e does not compromise the material's effectiveness in radiation shielding.

6.3. Half value layer (HVL) and mean free path (MFP)

Figs. 11 and 12 show the fluctuation of the half-value layer (HVL) and mean free path (MFP) with energy across the examined Bi0.0 – Bi0.45 glass samples.

The figures indicate a consistent pattern in both HVL and MFP concerning energy, with variations in magnitude. Initially, at energy 0.015 MeV, both HVL and MFP values were [0.021, 0.003, 0.002 and 0.001

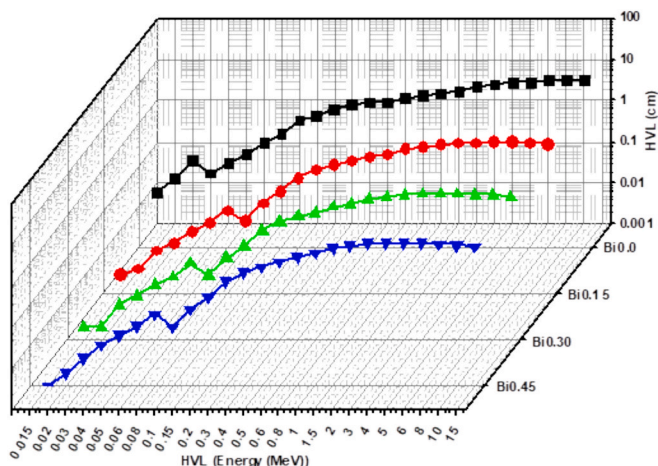


Fig. 11. Half value layer (HVL) values versus photon energy for the samples.

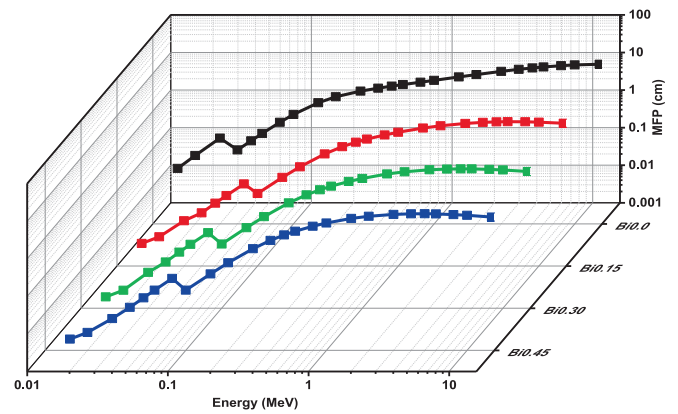


Fig. 12. Mean free path (MFP) values versus photon energy for the samples.

cm] and [0.030, 0.004, 0.002 and 0.002 cm] for Bi0.0-Bi0.45 samples, respectively. Subsequently, at energy higher than 1 MeV, the HVL and MFP witness an escalation with increasing energy levels. For example, the HVL for Bi0.0 at 1 MeV and 15 MeV records 4.567 cm and 12.239 cm, highlighting that higher photon energy results in decreased photon-sample interaction probability, allowing more photons to penetrate the sample.

Furthermore, Figs. 11 and 12 indicate an elevation in HVL and MFP of the prepared glasses as the Bi_2O_3 content increases. The inclusion of Bi_2O_3 enhances the glass density, elucidating the decline in HVL and MFP values with the incorporation of Bi_2O_3 .

Observing Figs. 11 and 12, there exists a progressive order of HVL and MFP from Bi0.0 to Bi0.45, with the glass ceramic sample Bi0.45 exhibiting the highest HVL and MFP values, while the Bi0.0 sample displays the lowest. This ranking can be attributed to the variable weight percentage of Bi_2O_3 . The superior gamma radiation shielding performance of Bi0.45 among the prepared glasses is highlighted by the HVL and MFP findings.

These results underscore the impact of bismuth oxide concentration on the materials' ability to attenuate gamma radiation, with higher concentrations leading to improved radiation shielding performance. Understanding the variations in HVL values across different energy levels provides essential information for evaluating the materials' effectiveness in reducing radiation exposure and enhancing safety in various applications.

The comparison of (HVL) values for the prepared glass and glass ceramic was conducted against established commercial glasses employed in gamma shielding, namely Rs-360 and Rs 253 G18, as well as ordinary and barium concrete (OC, BaC), and elements such as Cr and

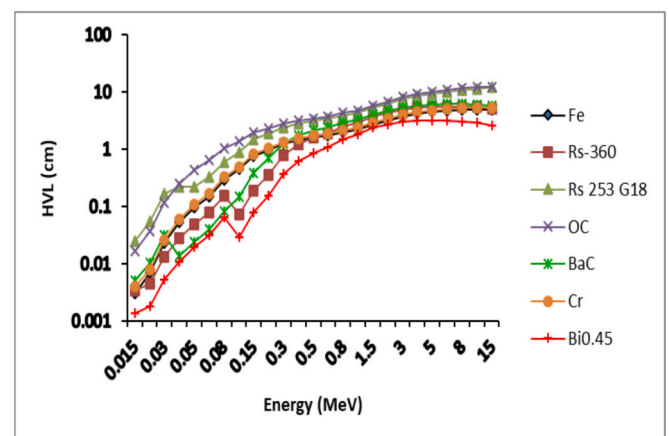


Fig. 13. A Comparison between HVL for Bi0.45 sample with concrete and commercial glasses.

Pb. The outcomes are illustrated in Fig. 13. Notably, the glass ceramic sample Bi0.45 exhibited a reduced HVL in comparison to all glasses and concrete counterparts. This outcome underscores the superior gamma ray shielding capability of sample Bi0.45 when compared with commercial glass and concrete materials.

6.4. Fast neutron absorption characteristics (FNAC)

The investigation into neutron protection within the current samples revealed notable neutron shielding capabilities, particularly in the Bi0.15 glass specimen. The substitution of the lighter oxide B_2O_3 with the heavier oxide Bi_2O_3 plays a significant role in influencing the neutron shielding properties of the glass samples. Boron oxide, with its relatively low atomic mass, primarily contributes differently to neutron moderation compared to heavy metal oxides such as bismuth oxide, which possess higher atomic numbers and densities. Heavier elements like bismuth have higher neutron absorption and scattering cross sections, enhancing the probability of interactions between neutrons and the glass matrix. This results in better attenuation of neutron radiation.

In this study, the Fast Neutron Removal Cross Section (FNAC) was evaluated for samples with varying Bi_2O_3 content, Fig. 14. The FNAC values increased from 0.093 for the undoped sample (Bi0.0) to 0.105 for the sample with 15 % Bi_2O_3 doping (Bi0.15), indicating an improvement in neutron shielding capacity with initial bismuth incorporation. However, further increase in Bi_2O_3 content to 30 % and 45 % (Bi0.30 and Bi0.45 samples) showed a slight decrease in FNAC values to 0.104 and 0.102 respectively. This trend may be attributed to structural modifications in the glass matrix, such as changes in network connectivity and packing density, that influence neutron moderation efficiency and the mean free path of neutrons (Kumar, 2019; Shu, 2018). Excessive Bi_2O_3 content can lead to phase separation or concentration quenching effects that adversely affect neutron shielding properties (Hussein, 2020).

Overall, these findings suggest that moderate doping of Bi_2O_3 optimizes neutron shielding effectiveness by enhancing neutron absorption, while excessive doping may slightly reduce this effect due to changes in the glass structure. This insight underscores the importance of balancing the content of lighter and heavier oxides to tailor the radiation shielding properties of glass materials. Upon comparison of the FNRC for all glass sample with standard materials like ordinary concrete (OC), graphite, and water, it was noted that Bi0.15 and Bi0.30 samples exhibited superior FNAC properties compared to these materials.

7. Conclusion

The glass samples were synthesized following the formulation: $(80-x) B_2O_3 + 10Na_2O + 10NaCl_2 + xBi_2O_3$, with x values of 0, 15, 30, and 45 mol%, utilizing the melt-quenching technique. X-ray diffraction analysis confirmed that samples Bi0.0 and Bi0.15 exhibited a glass structure, while samples Bi0.30 and Bi0.45 displayed a devitrified glass. The optical band gap ($E_{opt.}$) decreased from 2.40 eV to 2.08 eV with increasing Bi_2O_3 content due to non-bridging oxygen (NBO) ions and the weaker Bi-O bond. Concurrently, the refractive index (n) increased from 2.58 to 2.70 due to the higher polarizability of Bi^{3+} ($1.508 \text{ \AA}^3$) compared to B^{3+} ($0.002 \text{ \AA}^3$).

The effectiveness of shielding against gamma rays and fast neutrons was revealing that the bismuth oxide concentration increased the samples' density, mass attenuation coefficient, and effective atomic number, while decreasing half-value layer and mean free path values. Moreover, electron density values increased in the Bi0.0 and Bi0.15 samples but decreased in the Bi0.30 and Bi0.45 samples. The findings emphasized the superior nuclear radiation capabilities of the Bi0.45 sample compared to commercial glasses (Rs 253 G18, Rs 360) and concrete materials (OC and BaC). Significantly, the Bi0.15 sample exhibited the highest values for fast neutron removal cross-section, exceeding common neutron-moderating substances like water, graphene, and concrete. This emphasizes the outstanding shielding attributes of the samples

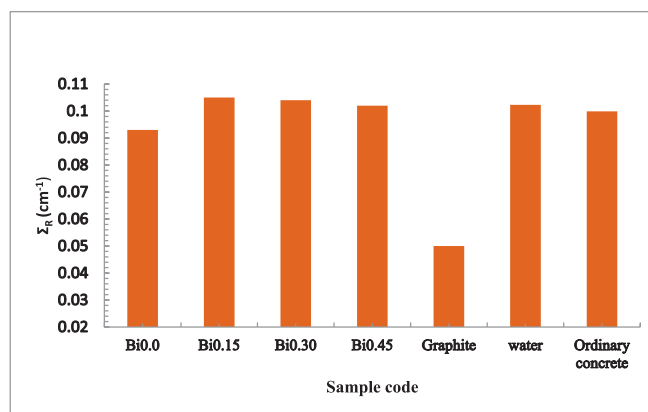


Fig. 14. Fast neutron removal cross section of the samples comparison with other materials.

developed in this research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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